Differentiating Integrals Under mild continuity restrictions, it is true that if

$$F(x) = \int_a^b g(t, x)\, dt,$$

then $F'(x) = \int_a^b g_x(t, x)\, dt$. Using this fact and the Chain Rule, we can find the derivative of

$$F(x) = \int_a^{f(x)} g(t, x)\, dt$$

by letting

$$G(u, x) = \int_a^u g(t, x)\, dt,$$

where $u = f(x)$. Find the derivatives of the functions in Exercises 51 and 52.

51. $F(x) = \int_0^{x^2} \sqrt{t^4 + x^3}\, dt$ **52.** $F(x) = \int_{x^2}^1 \sqrt{t^3 + x^2}\, dt$

Case I: $F(x) = \int_a^b g(t,x)dt$，即积分上下限是固定的数，与x无关，被积函数$g(t,x)$与x有关.

此时根据导数的定义:

$$F'(x) = \lim_{\Delta x \to 0} \frac{F(x+\Delta x) - F(x)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{\int_a^b g(t, x+\Delta x)dt - \int_a^b g(t,x)dt}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \int_a^b \frac{g(t,x+\Delta x) - g(t,x)}{\Delta x} dt$$

积分与极限交换顺序须满足某些连续条件，高数中都假设满足

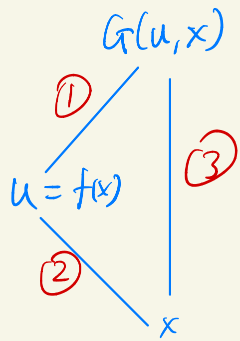
$$= \int_a^b \lim_{\Delta x \to 0} \frac{g(t,x+\Delta x) - g(t,x)}{\Delta x} dt$$

$$= \int_a^b \frac{\partial g(t,x)}{\partial x} dt = \int_a^b g_x'(t,x)dt$$

□

Case II: $F(x)=\int_a^{f(x)} g(t,x)\,dt$，即积分下限是固定的数，与 x 无关，积分上限为 $f(x)$，与 x 有关，被积函数 $g(t,x)$ 与 x 有关.

此时构造二元函数 $\boxed{G(u,x):=\int_a^u g(t,x)\,dt}$，那么 $F(x)=G(f(x),x)$.



那么

$$\frac{dF(x)}{dx}=\frac{dG(f(x),x)}{dx}$$

$$=①\cdot②+③$$

$$=\frac{\partial G}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial G}{\partial x}$$

$$=g(u,x)\cdot f'(x)+\int_a^u g'_x(t,x)\,dt$$

此时对 u 求偏导应把 x 视为常数，这是关于 u 的单变量积分变上限函数的求导

此时对 x 求偏导应把 u 视为常数，这就归化到 Case I 的求单变量导数的情况.

$$=g(f(x),x)\cdot f'(x)+\int_a^{f(x)} g'_x(t,x)\,dt \qquad (*)$$

□

Exercise: **Case III**: $F(x)=\int_{h(x)}^{f(x)} g(t,x)\,dt$ 如何对 x 求导?

Answer: 此时

$$F'(x)=g(f(x),x)\cdot f'(x)+\int_a^{f(x)} g'_x(t,x)\,dt$$

$$-g(h(x),x)\cdot h'(x)+\int_{h(x)}^{a} g'_x(t,x)\,dt$$

$$=g(f(x),x)f'(x)-g(h(x),x)h'(x)+\int_{h(x)}^{f(x)} g'_x(t,x)\,dt$$

□

Ex51 $F(x)=\int_0^{x^2}\sqrt{t^4+x^3}\,dt$

Sol: 对比 Case Ⅱ 知，此时 $\begin{cases} f(x)=x^2 \\ g(t,x)=\sqrt{t^4+x^3} \end{cases}$

套入公式 (*) 知：

$$F'(x)=g(f(x),x)\cdot f'(x)+\int_0^{f(x)} g'_x(t,x)\,dt$$

$$=\sqrt{(x^2)^4+x^3}\cdot 2x+\int_0^{x^2}\frac{3x^2}{2\sqrt{t^4+x^3}}\,dt$$

$$=2x^2\sqrt{x^6+x}+\frac{3x^2}{2}\int_0^{x^2}\frac{1}{\sqrt{t^4+x^3}}\,dt$$

$\square$

2024 Calculus II Additional Exercise (8)

1. Page 39, Exercise 1,2,3,4, and Page 40, Exercise 8 from Calculus Supplement Homework.

2. Exercise 33 (3),(4) in page 42 from Calculus Supplement Homework.

3. Exercise 43 in page 43 from Calculus Supplement Homework.

4. Exercise 45 in page 43 from Calculus Supplement Homework.

5. Exercise 47 in page 43 from Calculus Supplement Homework.

Q1、P39 Ex1,2,3,4.

1. Define $f(x,y) = \frac{xy}{x^2+y^2}$ when $(x,y) \neq (0,0)$ and $f(x,y) = 0$ when $(x,y) = (0,0)$. Then at $(0,0)$, $f(x,y)$ is C
   (A) continuous, and partial derivatives exist.
   (B) continuous, but partial derivatives do not exist.
   (C) not continuous, but partial derivatives exist.
   (D) not continuous, and partial derivatives do not exist.
2. Assume $f(x,y) = y(x-1)^2 + x(y-2)^2$. Which of the following method computing $f_x(1,2)$ is incorrect ? B
   (A) Since $f(x,2) = 2(x-1)^2$, $f_x(x,2) = 4(x-1)$, $f_x(1,2) = 0$.
   (B) Since $f(1,2) = 0$, $f_x(1,2) = 0' = 0$.
   (C) Since $f_x(x,y) = 2y(x-1) + (y-2)^2$, $f_x(1,2) = 0$.
   (D) $f_x(1,2) = \lim\limits_{x\to 1} \dfrac{f(x,2) - f(1,2)}{x-1} = 0$.
3. Suppose both partial derivatives for $f(x,y)$ exist at the point (x_0, y_0). Then C
   (A) $f(x,y)$ is bounded near (x_0, y_0).
   (B) $f(x,y)$ is continuous near (x_0, y_0).
   (C) $f(x,y_0)$ is continuous at x_0, $f(x_0,y)$ is continuous at y_0.
   (D) $f(x,y)$ is continuous at (x_0, y_0).
4. Let $x = x(y,z)$, $y = y(z,x)$ and $z = z(x,y)$ are implicit function determined by $F(x,y,z) = 0$. Then which of the following is incorrect? C
   (A) $\frac{\partial x}{\partial y}\frac{\partial y}{\partial x} = 1$.
   (B) $\frac{\partial x}{\partial z}\frac{\partial z}{\partial x} = 1$.
   (C) $\frac{\partial x}{\partial y}\frac{\partial y}{\partial z}\frac{\partial z}{\partial x} = 1$.
   (D) $\frac{\partial x}{\partial y}\frac{\partial y}{\partial z}\frac{\partial z}{\partial x} = -1$.

Ex1

$$f_x'(0,0) = \lim_{x\to 0} \frac{f(x,0)-f(0,0)}{x} = \lim_{x\to 0} 0 = 0$$

Ex3 $f(x,y)$ bounded near (x_0,y_0) means: $\exists$ (x_0,y_0) 的邻域 U, $\exists M>0$, s.t. $\forall (x,y)\in U$, 有 $|f(x,y)| < M$.

特别地, $f(x,y)$ 在 (x_0,y_0) 处连续 $\Rightarrow$ $f(x,y)$ 在 (x_0,y_0) 附近有界

连续性 $\Rightarrow$ for $\forall \varepsilon > 0$, $\exists \delta$, s.t. w $\sqrt{(x-x_0)^2+(y-y_0)^2} < \delta$, $|f(x,y)-f(x_0,y_0)| < \varepsilon$

现在令 $U = \mathbb{B}((x_0,y_0),\delta)$，即以 (x_0,y_0) 为圆心，δ 为半径的邻域，

令 $M = |f(x_0,y_0)|+1$，则 $\forall (x,y) \in U$,

$$|f(x,y)| - |f(x_0,y_0)| \le |f(x,y)-f(x_0,y_0)| < \varepsilon = 1$$

$$\Rightarrow \quad |f(x,y)| < 1 + |f(x_0,y_0)|$$

所以该题 (B) 对 $\Rightarrow$ (D) 对 $\Rightarrow$ (A) 对，但是 (A) 是不对的！

反例：
$$f(x,y) = \begin{cases} 1 & x=0 \text{ or } y=0 \\ \dfrac{1}{xy} & \text{otherwise} \end{cases}$$

Ex 4 在 $F(x,y,z)=0$ 中，$x\,y\,z$ 有两个是独立变量！

(1) x, y 独立，即 $z = z(x,y)$

那么 $F(x,y,z(x,y))=0$

$$\begin{cases} F'_x + F'_z \cdot \dfrac{\partial z}{\partial x} = 0 \\ F'_y + F'_z \dfrac{\partial z}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} \dfrac{\partial z}{\partial x} = -\dfrac{F'_x}{F'_z} \\ \dfrac{\partial z}{\partial y} = -\dfrac{F'_y}{F'_z} \end{cases}$$

(2) y, z 独立，即 $x = x(y,z)$

那么 $F(x(y,z), y, z) = 0$

$$\begin{cases} F'_x \cdot \dfrac{\partial x}{\partial y} + F'_y = 0 \\ F'_x \dfrac{\partial x}{\partial z} + F'_z = 0 \end{cases} \Rightarrow \begin{cases} \dfrac{\partial x}{\partial y} = -\dfrac{F'_y}{F'_x} \\ \dfrac{\partial x}{\partial z} = -\dfrac{F'_z}{F'_x} \end{cases}$$

(3) x, z 独立，即 $y = y(x,z)$

那么 $F(x, y(x,z), z) = 0$

$$\begin{cases} F'_x + F'_y \dfrac{\partial y}{\partial x} = 0 \\ F'_z + F'_y \dfrac{\partial y}{\partial z} = 0 \end{cases} \Rightarrow \begin{cases} \dfrac{\partial y}{\partial x} = \dfrac{-F'_x}{F'_y} \\ \dfrac{\partial y}{\partial z} = \dfrac{-F'_z}{F'_y} \end{cases}$$

P40 Ex8

8. Suppose f is continuous at $(0,0)$. Then which of the following statement is correct ?

(A) $\lim_{(x,y)\to(0,0)} \frac{f(x,y)}{|x|+|y|}$ exists, then f is differentiable at $(0,0)$.

(B) $\lim_{(x,y)\to(0,0)} \frac{f(x,y)}{x^2+y^2}$ exists, then f is differentiable at $(0,0)$.

(C) f is differentiable at $(0,0)$, then $\lim_{(x,y)\to(0,0)} \frac{f(x,y)}{|x|+|y|}$ exists.

(D) f is differentiable at $(0,0)$, then $\lim_{(x,y)\to(0,0)} \frac{f(x,y)}{x^2+y^2}$ exists.

B

Sol：反例：(A) $f(x,y)=|x|+|y|$

(C), (D), $f(x,y)\equiv 1$

(B): $\lim\limits_{(x,y)\to(0,0)} \frac{f(x,y)}{x^2+y^2}=L \Rightarrow f(0,0)=\lim\limits_{(x,y)\to(0,0)} f(x,y)=\lim\limits_{(x,y)\to(0,0)} \frac{f(x,y)}{x^2+y^2}(x^2+y^2)$

$=0$

along $y=0$: $\lim\limits_{x\to 0} \frac{f(x,0)}{x^2}=L$

Then $\frac{\partial f}{\partial x}(0,0)=\lim\limits_{x\to 0} \frac{f(x,0)-f(0,0)}{x-0}$

$=\lim\limits_{x\to 0} \frac{f(x,0)}{x^2}\cdot x$

$=0$

同理 $\frac{\partial f}{\partial y}(0,0)=0$

计算 $\Delta z - f'_x(0,0)\Delta x - f'_y(0,0)\Delta y$

$=f(\Delta x,\Delta y)$

$=\underbrace{\left(\frac{f(\Delta x,\Delta y)\Delta x}{(\Delta x)^2+\Delta y^2}\right)}_{\varepsilon_1}\Delta x+\underbrace{\left(\frac{f(\Delta x,\Delta y)}{(\Delta x)^2+(\Delta y)^2}\right)}_{\varepsilon_2}\Delta y$

$\lim\limits_{(\Delta x,\Delta y)\to(0,0)} \varepsilon_1=\lim\limits_{(\Delta x,\Delta y)\to(0,0)} \varepsilon_2=0 \Rightarrow f(x,y)$ 在 $(0,0)$ 可微.

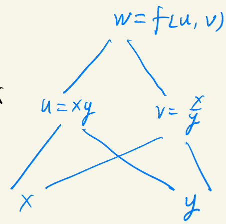
Q2 P42 Ex33 (3)(4)

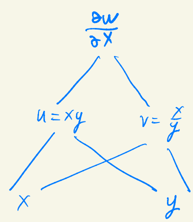
33. Calculating all second order partial derivatives:

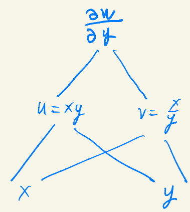
(1) $f(x,y) = \arctan\dfrac{x+y}{1-xy}$; (2) $f(x,y) = \ln\left(x+y^2\right)$;

(3) $w = f\left(xy, \dfrac{x}{y}\right)$; (4) $z = f\left(xy, x^2+y^2\right)$.

Sol: (3)







$$f'_x = f'_u \cdot u'_x + f'_v \cdot v'_x = y \cdot f'_u + \frac{1}{y} \cdot f'_v$$

$$f'_y = f'_u \cdot u'_y + f'_v \cdot v'_y = x \cdot f'_u - \frac{x}{y^2} f'_v$$

$$f''_{xx} = \left(y \cdot f'_u\right)'_x + \left(\frac{1}{y} \cdot f'_v\right)'_x = y\left(f''_{uu} u'_x + f''_{uv} v'_x\right) + \frac{1}{y}\left(f''_{vu} u'_x + f''_{vv} v'_x\right)$$

$$= y\left(f''_{uu} y + f''_{uv} \cdot \frac{1}{y}\right) + \frac{1}{y}\left(f''_{vu} y + f''_{vv} \cdot \frac{1}{y}\right)$$

$$= y^2 f''_{uu} + f''_{uv} + f''_{vu} + \frac{1}{y^2} f''_{vv}$$

$$f''_{xy} = \left(y \cdot f'_u\right)'_y + \left(\frac{1}{y} \cdot f'_v\right)'_y$$

$$= f'_u + y\left(f''_{uu} u'_y + f''_{uv} \cdot v'_y\right) + \left(-\frac{f'_v}{y^2}\right) + \frac{1}{y}\left(f''_{vu} \cdot u'_y + f''_{vv} v'_y\right)$$

$$= f'_u + y\left(f''_{uu} x + f''_{uv} \cdot \left(-\frac{x}{y^2}\right)\right) - \frac{f'_v}{y^2} + \frac{1}{y}\left(f''_{vu} x + f''_{vv}\left(-\frac{x}{y^2}\right)\right)$$

$$= f'_u - \frac{f'_v}{y^2} + xy f''_{uu} - \frac{x}{y} f''_{uv} + \frac{x}{y} f''_{vu} - \frac{x}{y^3} f''_{vv}$$

$$f''_{yx}=(x\cdot f'_u)'_x-\left(\frac{x}{y^2}f'_v\right)'_x$$

$$=f'_u+x(f''_{uu}\cdot u'_x+f''_{uv}\cdot v'_x)-\frac{1}{y^2}f'_v-\frac{x}{y^2}(f''_{vu}\cdot u'_x+f''_{vv}\cdot v'_x)$$

$$=f'_u+x\left(f''_{uu}\cdot y+f''_{uv}\cdot\frac{1}{y}\right)-\frac{1}{y^2}f'_v-\frac{x}{y^2}\left(f''_{vu}\cdot y+f''_{vv}\cdot\frac{1}{y}\right)$$

$$=f'_u-\frac{1}{y^2}f'_v+xyf''_{uu}+\frac{x}{y}f''_{uv}-\frac{x}{y}f''_{vu}-\frac{x}{y^3}f''_{vv}$$

$$f''_{yy}=(x\cdot f'_u)'_y-\left(\frac{x}{y^2}f'_v\right)'_y$$

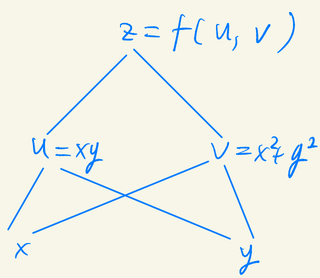
$$=x(f''_{uu}\cdot u'_y+f''_{uv}\cdot v'_y)+\frac{2x}{y^3}f'_v-\frac{x}{y^2}(f''_{vu}\cdot u'_y+f''_{vv}\cdot v'_y)$$

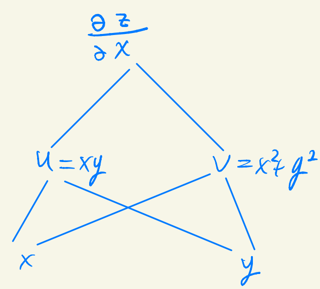
$$=x\left(f''_{uu}\cdot x+f''_{uv}\cdot\left(-\frac{x}{y^2}\right)\right)+\frac{2x}{y^3}f'_v-\frac{x}{y^2}\left(f''_{vu}\cdot x+f''_{vv}\cdot\left(-\frac{x}{y^2}\right)\right)$$

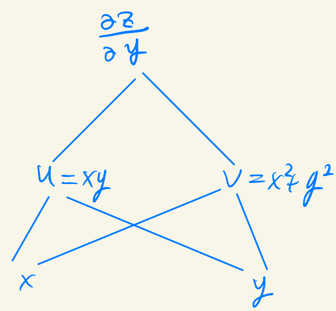
$$=\frac{2x}{y^3}f'_v+x^2f''_{uu}-\frac{x^2}{y^2}f''_{uv}-\frac{x^2}{y^2}f''_{vu}+\frac{x^2}{y^4}f''_{vv}$$

(4) 换一种写法：已知

$$\begin{cases}\dfrac{\partial u}{\partial x}=y\\ \dfrac{\partial u}{\partial y}=x\\ \dfrac{\partial v}{\partial x}=2x\\ \dfrac{\partial v}{\partial y}=2y\end{cases}$$







$$\therefore\ \frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}=y\frac{\partial z}{\partial u}+2x\frac{\partial z}{\partial v}$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}=x\frac{\partial z}{\partial u}+2y\frac{\partial z}{\partial v}$$

二阶导们;

$$\frac{\partial^2 z}{\partial x^2}=\frac{\partial\left[y\frac{\partial z}{\partial u}+2x\frac{\partial z}{\partial v}\right]}{\partial x}$$

$$=y\left(\frac{\partial^2 z}{\partial u^2}\cdot\frac{\partial u}{\partial x}+\frac{\partial^2 z}{\partial v\partial u}\cdot\frac{\partial v}{\partial x}\right)+2\frac{\partial z}{\partial v}+2x\left(\frac{\partial z^2}{\partial u\partial v}\cdot\frac{\partial u}{\partial x}+\frac{\partial^2 z}{\partial v^2}\cdot\frac{\partial v}{\partial x}\right)$$

$$=y\left(\frac{\partial^2 z}{\partial u^2}\cdot y+\frac{\partial^2 z}{\partial v\partial u}\cdot(2x)\right)+2\frac{\partial z}{\partial v}+2x\left(\frac{\partial^2 z}{\partial u\partial v}\cdot y+\frac{\partial^2 z}{\partial v\partial u}\cdot(2x)\right)$$

$$=2\frac{\partial z}{\partial v}+y^2\frac{\partial^2 z}{\partial u^2}+2xy\frac{\partial^2 z}{\partial v\partial u}+2xy\frac{\partial^2 z}{\partial u\partial v}+4x^2\frac{\partial^2 z}{\partial v\partial u}$$

$$\frac{\partial^2 z}{\partial y\partial x}=\frac{\partial\left[y\frac{\partial z}{\partial u}+2x\frac{\partial z}{\partial v}\right]}{\partial y}$$

$$=\frac{\partial z}{\partial u}+y\left[\frac{\partial^2 z}{\partial u^2}\cdot\frac{\partial u}{\partial y}+\frac{\partial^2 z}{\partial v\partial u}\cdot\frac{\partial v}{\partial y}\right]+2x\left[\frac{\partial^2 z}{\partial u\partial v}\cdot\frac{\partial u}{\partial y}+\frac{\partial^2 z}{\partial v^2}\frac{\partial v}{\partial y}\right]$$

$$=\frac{\partial z}{\partial u}+xy\frac{\partial^2 z}{\partial u^2}+2y^2\frac{\partial^2 z}{\partial v\partial u}+2x^2\frac{\partial^2 z}{\partial u\partial v}+4xy\frac{\partial^2 z}{\partial v^2}$$

$$\frac{\partial^2 z}{\partial x\partial y}=\frac{\partial\left[x\frac{\partial z}{\partial u}+2y\frac{\partial z}{\partial v}\right]}{\partial x}$$

$$=\frac{\partial z}{\partial u}+x\left[\frac{\partial^2 z}{\partial u^2}\cdot\frac{\partial u}{\partial x}+\frac{\partial^2 z}{\partial v\partial u}\cdot\frac{\partial v}{\partial x}\right]+2y\left[\frac{\partial^2 z}{\partial u\partial v}\cdot\frac{\partial u}{\partial x}+\frac{\partial^2 z}{\partial v^2}\cdot\frac{\partial v}{\partial x}\right]$$

$$=\frac{\partial z}{\partial u}+xy\frac{\partial^2 z}{\partial u^2}+2x^2\frac{\partial^2 z}{\partial v\partial u}+2y^2\frac{\partial^2 z}{\partial u\partial v}+2x\frac{\partial^2 z}{\partial v^2}$$

$$\frac{\partial^2 z}{\partial y^2}=\frac{\partial\left[x\frac{\partial z}{\partial u}+2y\frac{\partial z}{\partial v}\right]}{\partial y}$$

$$=x\left[\frac{\partial^2 z}{\partial u^2}\cdot\frac{\partial u}{\partial y}+\frac{\partial^2 z}{\partial v\partial u}\cdot\frac{\partial v}{\partial y}\right]+2\frac{\partial z}{\partial v}+2y\left[\frac{\partial^2 z}{\partial u\partial v}\cdot\frac{\partial u}{\partial y}+\frac{\partial^2 z}{\partial v^2}\cdot\frac{\partial v}{\partial y}\right]$$

$$=2\frac{\partial z}{\partial v}+x^2\frac{\partial^2 z}{\partial u^2}+2xy\frac{\partial^2 z}{\partial v\partial u}+2xy\frac{\partial^2 z}{\partial u\partial v}+4y^2\frac{\partial^2 z}{\partial v^2}$$

Q3 P43 Ex43

43. Let

$$f(x,y)=\begin{cases} y\sin\dfrac{1}{x^2+y^2}, & (x,y)\neq(0,0);\\ 0, & (x,y)=(0,0).\end{cases}$$

(1) Find f_x and f_y for any point other than the origin;

(2) Determine if $f_x(0,0)$ and $f_y(0,0)$ exist or not;

(3) Find all points at which f is continuous.

Sol: (1) $f_x = -\dfrac{2xy}{(x^2+y^2)^2}\cos\dfrac{1}{x^2+y^2}$

$f_y = \sin\dfrac{1}{x^2+y^2} - \dfrac{2y^2}{(x^2+y^2)^2}\cos\dfrac{1}{x^2+y^2}$

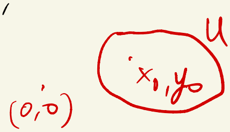
(2) $f_x(0,0) = \lim\limits_{x\to0}\dfrac{f(x,0)-f(0,0)}{x-0} = 0$

$f_y(0,0) = \lim\limits_{y\to0}\dfrac{f(0,y)-f(0,0)}{y-0} = \lim\limits_{y\to0}\dfrac{y\sin\frac{1}{y^2}}{y}$ 不存在

(3) ① 所有 f 在 $\mathbb{R}^2\setminus\{(0,0)\}$ 连续!

Reason: $\forall (x_0,y_0)\neq(0,0)$, $\exists (x_0,y_0)$ 的邻域, s.t. $(0,0)\notin U$

则 $f(x,y)=y\sin\dfrac{1}{x^2+y^2}$ on U



$$\therefore \lim_{(x,y)\to(x_0,y_0)} f(x,y) = \lim_{(x,y)\to(x_0,y_0)} y\sin\frac{1}{x^2+y^2} = y_0\sin\frac{1}{x_0^2+y_0^2}$$

$$= f(x_0,y_0).$$

② 考虑 $(0,0)$ 处连续性：

$$0 \le \left| y\sin\frac{1}{x^2+y^2} \right| \le |y|$$

$(x,y)\to(0,0)$ ↓ $0=f(0,0)$ ↓ $(x,y)\to(0,0)$ 0

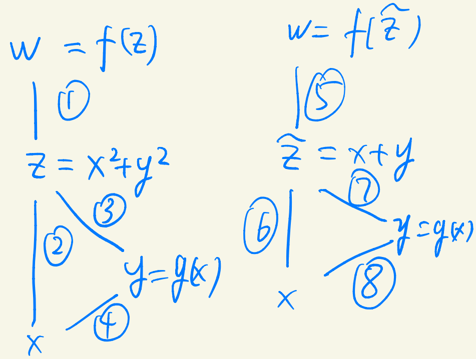
$\therefore f(x,y)$ 在 $\mathbb{R}^2$ 上连续

$\square$

Q4 P43 Ex45

45. Suppose that $w = f(z)$ is a differentiable function and the equation $f(x^2+y^2)+f(x+y) = y$ defines a differentiable implicit function $y = g(x)$. Given $g(0) = 2$, $f'(2) = 0.5$ and $f'(4) = 1$, find $g'(0)$.

Sol:



方程 $f(x^2+y^2)+f(x+y)=y$ 两边对 x 求导:

$$f'\cdot(2x+2y\cdot g')+f'\cdot(1+g')=g'$$

代入 $\begin{cases} x=0 \\ y=g(0)=2 \end{cases}$ 得 $f'(4)\cdot 4g'(0)+f'(2)\cdot(1+g'(0))=g'(0)$

$$4g'(0)+\frac{1}{2}(1+g'(0))=g'(0)$$

$$g'(0)=\frac{-1}{7}$$

Q5 P43 Ex47

47. Determine if the following function is continuous at $(0,0)$ and state your reasons.

$$f(x,y) = \begin{cases} \dfrac{x^2\sqrt{|y|}}{x^2+y^2}, & (x,y) \neq (0,0); \\ 0, & (x,y) = (0,0). \end{cases}$$

Sol:

$$0 \leq \frac{x^2\sqrt{|y|}}{x^2+y^2} \leq \frac{x^2\sqrt{|y|}}{x^2} = \sqrt{|y|}$$

$$\underset{0}{\downarrow} \qquad \underset{0}{\downarrow}{\scriptstyle (x,y)\to(0,0)} \qquad \underset{0}{\downarrow}{\scriptstyle (x,y)\to(0,0)}$$

$$\therefore \lim_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0)$$

Then $f(x,y)$ continuous at $(0,0)$.

$\square$